Resource Estimates for Nuclear Lattice Effective Field Theories

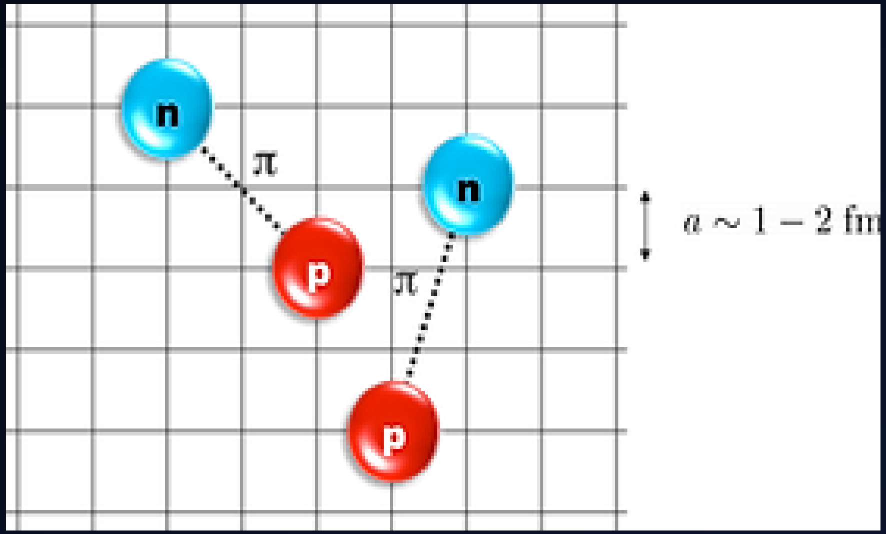
REDUCING RESOURCES BY IMPLEMENTING QUBITIZATION

BRANDON FRIEND AND TRIET (TONY) HA

Contents

Theory: Nuclear Lattice Effective Field Theories (EFTs)

- "Chiral EFTs" on a lattice – a theory for low-energy nuclear physics
- **Dynamical-Pion EFT** where long-range interactions between nucleons (fermions) are mediated by the pion field (bosons)
- **The mix of fermionic and bosonic operators makes this a challenging problem**



Lattice Effective Field Theory, Lee Research Group, FRIB, https://www.lee-research-group.com/lattice-effective-field-theory

Theory: Dynamical Pion EFT Hamiltonian

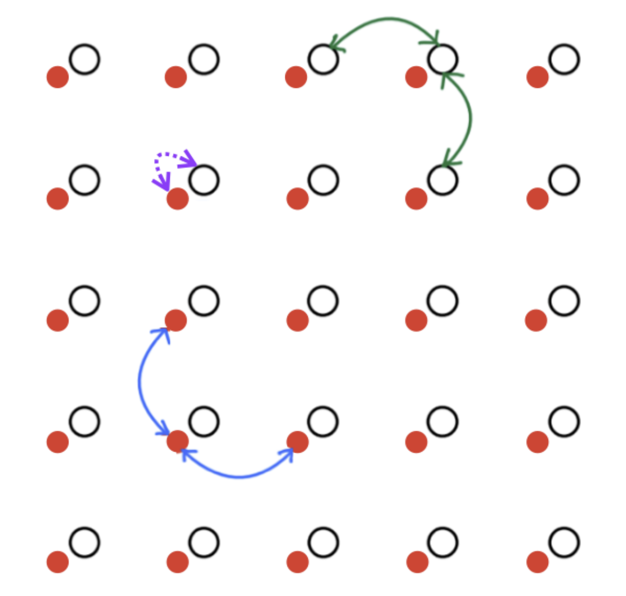
$$H_{D\pi} = \underbrace{H_{\text{free}} + H_C + H_{C_{I^2}}}_{\text{fermionic}} + \underbrace{H_\pi + H_{N\pi}}_{\text{bosonic}}$$

H_{free} Free nucleon term

H_C $H_{C_{I^2}}$ Short-range Nucleon-Nucleon interactions

H_π Free pion term

$H_{N\pi}$ Nucleon-Pion Interaction


Dynamical-Pion EFT

Theory: Dynamical Pion EFT Hamiltonian

$$H_{D\pi} = H_{\text{free}} + H_C + H_{C_{I^2}} + H_\pi + H_{N\pi}$$

Degree of freedoms

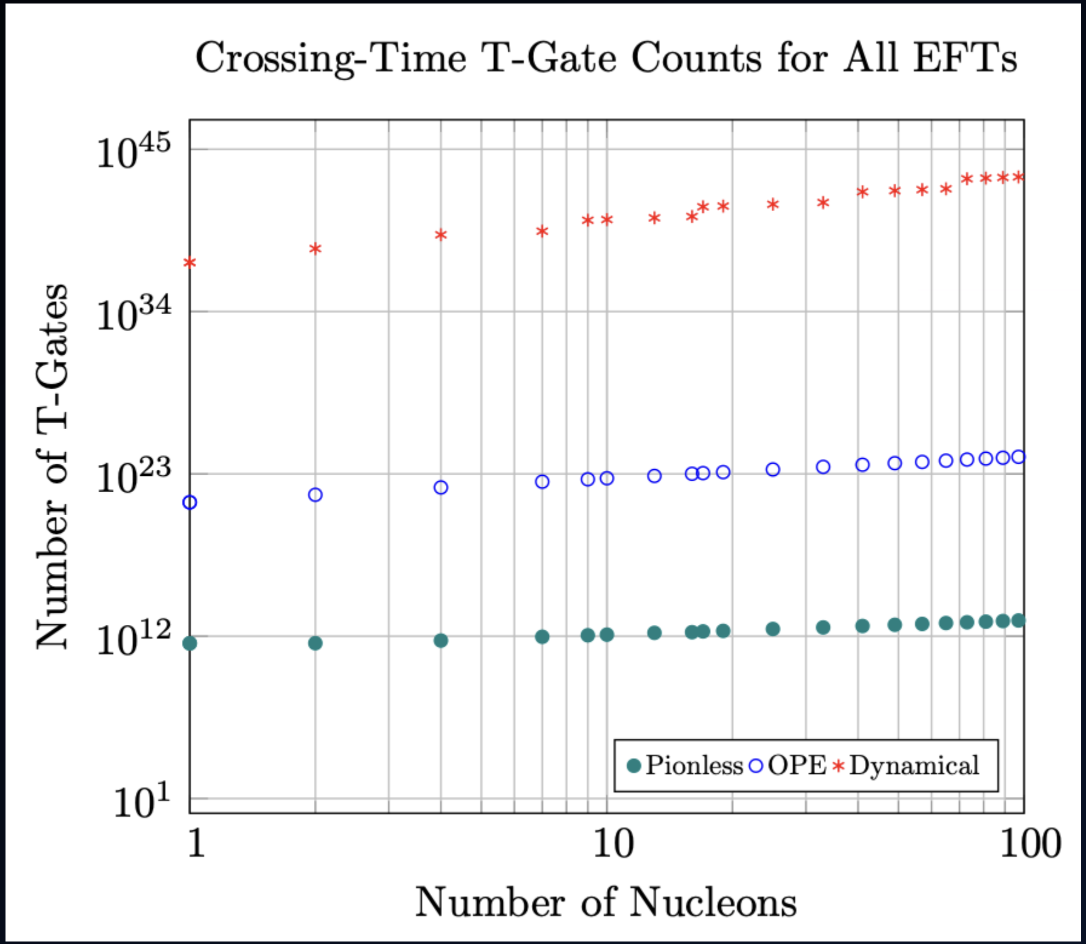
- L^3 sites for 3D lattice
- Three pion flavors π^0, π^+, π^-
- Two spins, two iso-spins
- Pion field cutoffs π_{max}, Π_{max}

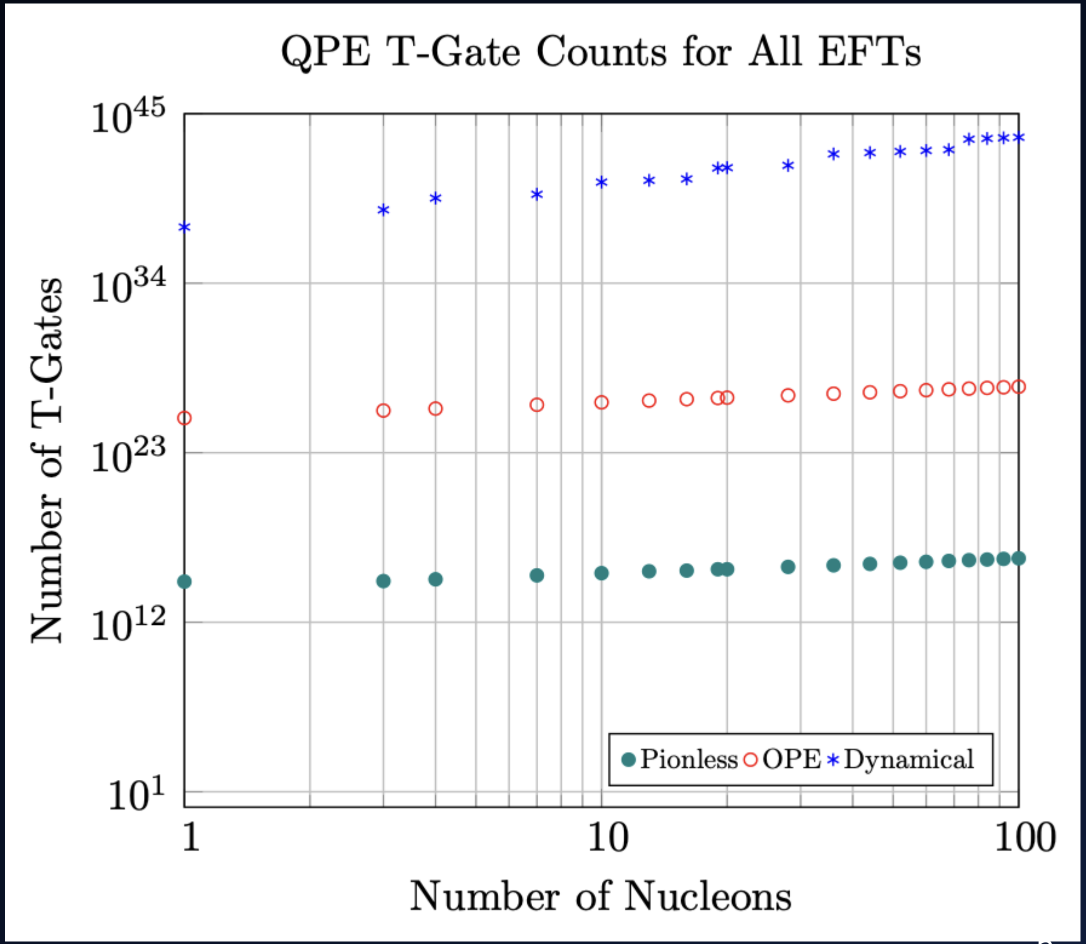
$$H_{N\pi} = H_{AV} + H_{WT}$$

$$H_{\text{WT}} = \frac{1}{4f_\pi^2} \sum_{\mathbf{x}} \sum_{I_1, I_2, I_3} \sum_{\alpha,\beta,\delta} \epsilon_{I_1 I_2 I_3}\, \pi_{I_2}(\mathbf{x})\, \Pi_{I_3}(\mathbf{x})\, a^\dagger_{\alpha\beta}(\mathbf{x}) [\tau_{I_1}]_{\beta\delta} a_{\alpha\delta}(\mathbf{x})$$

Quantum Algorithms for Simulating Nuclear Effective Field Theories

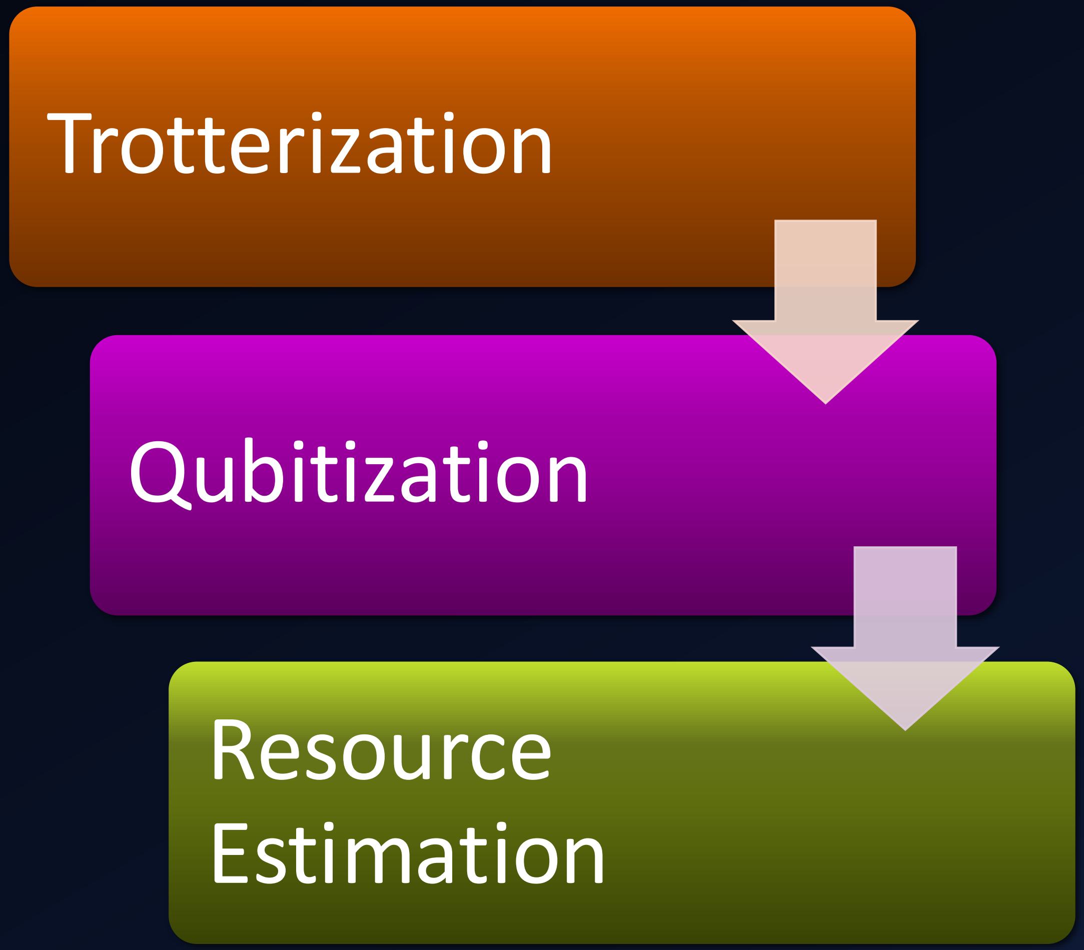
James D. Watson,[1,2] Jacob Bringewatt,[3,1,4,5] Alexander F. Shaw,[1,5]
Andrew M. Childs,[1,2] Alexey V. Gorshkov,[1,4] and Zohreh Davoudi[1,5,6]





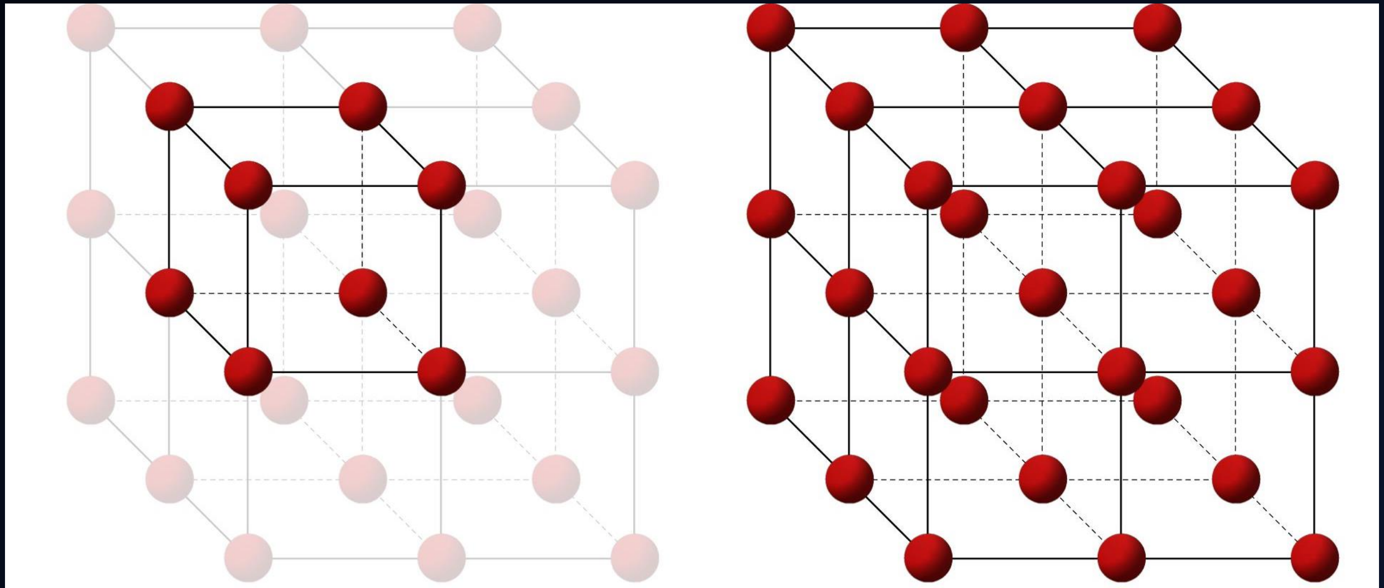
Project: Implement Qubitization

- Previous work performed resource estimation for Chiral EFTs including Dynamical Pion EFT using Trotterization as a subroutine (Watson, et. al.)
- Our project was to implement qubitization as the subroutine for evaluating the time evolution operator e^{iHt} and compare the resources



Project: Resource Scaling with System Size

- We investigated how resource costs scale with system size:
  - Lattice sites per side (L)
  - Total nucleons (A)

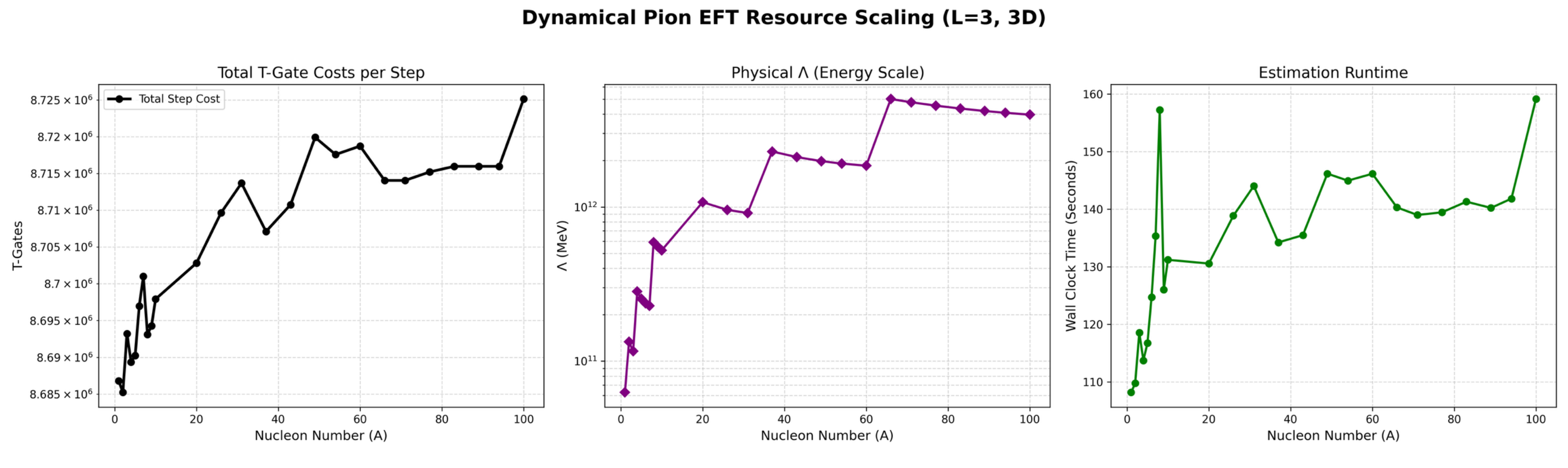


Simple cubic lattice structure (unit cell), Tec-Science, https://www.tec-science.com/material-science/structure-of-metals/important-types-of-lattice-structures/

Methods: OpenFermion and pyLIQTR

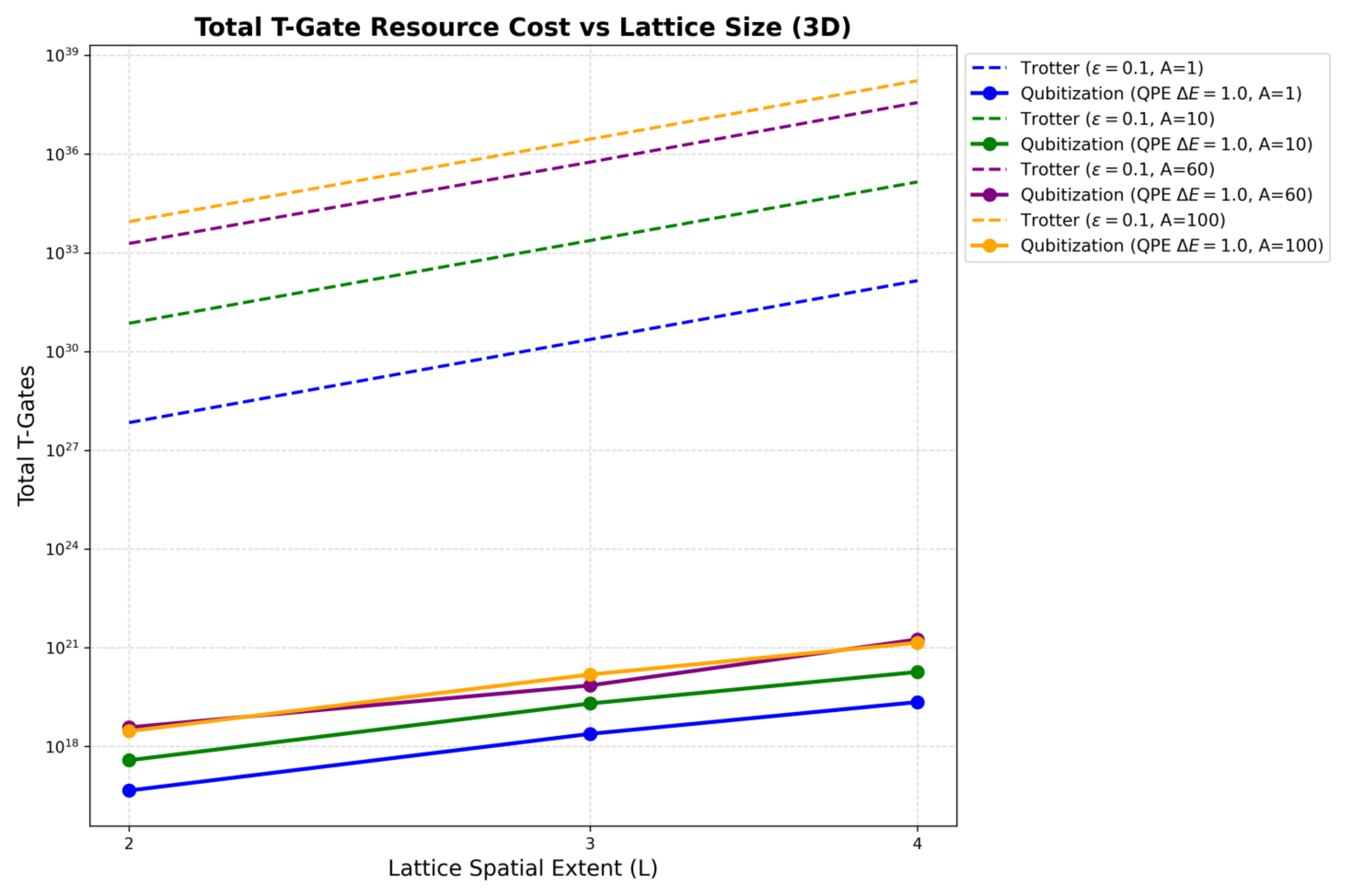
- We used the encoding for the pion field and values of the coefficients in the Hamiltonian from Watson, et. al.
- Constructed the Hamiltonian using OpenFermion and the Jordan-Wigner Transformation
- Estimated the logical resources necessary to implement a walk operator using pyLIQTR

Results



- For a given lattice size L, we constructed the Hamiltonian and estimated the resources for various number of nucleons (A)
- Complexity depends greatly on the normalization constant Λ

Results



- We used the bounds derived in Watson, et. al. to estimate Trotter resources
- We used the pyLIQTR resource estimates for the qubitization curves

Discussion

- Qubitization showed a significantly lower total cost in T-gates than Trotterization
- Resources scale much better with lattice size L and nucleon number A for qubitized walk operators than for Trotterized evolution operators
- The time to construct the Hamiltonian and estimate the resources grew quickly with lattice size, so we were limited to only L=2, 3, & 4

Future work

- Compare logical qubit requirements between qubitization and Trotterization
- We plan to optimize the code and run for larger lattices using high performance computing
- Watson, et. al. used the Verstraete-Cirac encoding for fermionic operators to reduce operator weight. We plan to study how impactful this is to the resource estimates

References

[1] R Machleidt and F Sammarruca. Chiral eft based nuclear forces: achievements and challenges. Physica Scripta, 91(8):083007, July 2016.

[2] Siddharth Hariprakash, Neel S. Modi, Michael Kreshchuk, Christopher F. Kane, and Christian W. Bauer. Strategies for simulating the time evolution of hamiltonian lattice field theories. Phys. Rev. A, 111:022419, Feb 2025.

[3] James D. Watson, Jacob Bringewatt, Alexander F. Shaw, Andrew M. Childs, Alexey V. Gorshkov, and Zohreh Davoudi. Quantum algorithms for simulating nuclear effective field theories, 2025.

[4] Jarrod R. McClean, et al., Openfermion: The electronic structure package for quantum computers, 2019.

[5] Guang Hao Low and Isaac L. Chuang. Hamiltonian Simulation by Qubitization. Quantum, 3:163, July 2019.

[6] Ryan Babbush, Craig Gidney, Dominic W. Berry, Nathan Wiebe, Jarrod McClean, Alexandru Paler, Austin Fowler, and Hartmut Neven. Encoding electronic spectra in quantum circuits with linear t complexity. Physical Review X, 8(4), October 2018.

Theory: Dynamical Pion EFT Hamiltonian

$$H_{D\pi} = H_{\text{free}} + H_C + H_{C_{I^2}} + H_\pi + H_{N\pi}$$

where the fermionic terms are the free nucleon term:

$$H_{\text{free}} = -h \sum_{\langle \mathbf{x},\mathbf{y}\rangle} \sum_{\sigma} \left(a_\sigma^\dagger(\mathbf{x}) a_\sigma(\mathbf{y}) + a_\sigma^\dagger(\mathbf{y}) a_\sigma(\mathbf{x}) \right) + 6h \sum_{\mathbf{x}} \sum_{\sigma} N_\sigma(\mathbf{x})$$

and the contact potentials for nucleons on the same site:

$$H_C = \frac{C}{2} \sum_{\mathbf{x}} \left[\sum_{\alpha} \sum_{\beta} a_{\alpha\beta}^\dagger(\mathbf{x})\, a_{\alpha\beta}(\mathbf{x}) \right]^2$$

$$H_{C_{I^2}} = \frac{C_{I^2}}{2} \sum_{\mathbf{x}} \sum_{I} : \left[\sum_{\alpha} \sum_{\beta,\delta} a_{\alpha\beta}^\dagger(\mathbf{x}) [\tau_I]_{\beta\delta}\, a_{\alpha\delta}(\mathbf{x}) \right]^2$$

Theory: Dynamical Pion EFT Hamiltonian

$$H_{D\pi} = H_{\text{free}} + H_C + H_{C_{I^2}} + H_\pi + H_{N\pi}$$

The pion field has additional terms including the free pion:

$$H_\pi = \frac{a_L^3}{2} \sum_{\mathbf{x}} \sum_I \left[\Pi_I^2(\mathbf{x}) + \left(\nabla \pi_I(\mathbf{x})\right)^2 + m_\pi^2 \pi_I^2(\mathbf{x}) \right]$$

And nucleon-pion interactions

$$H_{N\pi} = H_{AV} + H_{WT}$$

$$H_{\text{AV}} = \frac{g_A}{2f_\pi} \sum_{\mathbf{x}} \sum_{\alpha,\beta,\gamma,\delta} \sum_{I,S} a^\dagger_{\alpha\beta}(\mathbf{x}) [\tau_I]_{\beta\delta} [\sigma_S]_{\alpha\gamma} \, \partial_S \pi_I(\mathbf{x}) \, a_{\gamma\delta}(\mathbf{x})$$

$$H_{\text{WT}} = \frac{1}{4f_\pi^2} \sum_{\mathbf{x}} \sum_{I_1,I_2,I_3} \sum_{\alpha,\beta,\delta} \epsilon_{I_1 I_2 I_3} \, \pi_{I_2}(\mathbf{x}) \, \Pi_{I_3}(\mathbf{x}) \, a^\dagger_{\alpha\beta}(\mathbf{x}) [\tau_{I_1}]_{\beta\delta} a_{\alpha\delta}(\mathbf{x})$$